Representation Learning for LHC Triggers

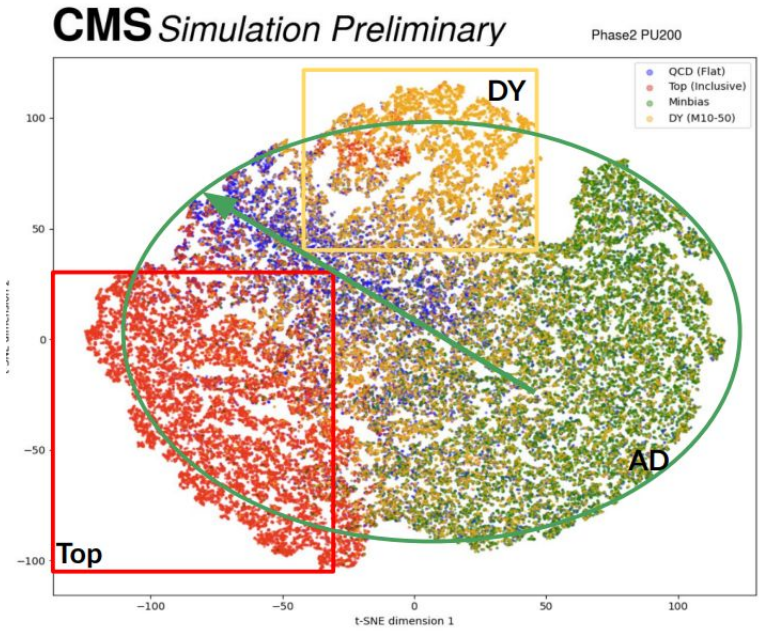
Estefania Moreno León on behalf of the
NextGen WP3 OpenLab

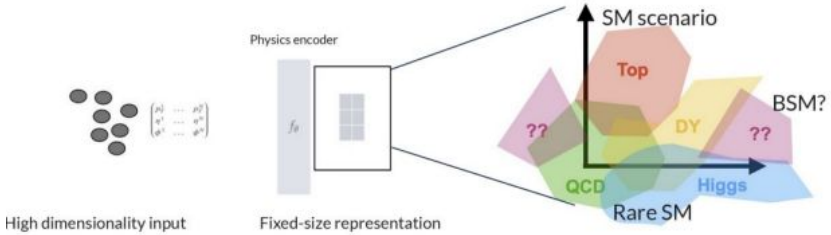
CERN | NexTGen Next Generation Triggers

Motivation & Research Questions

NGT WP3 investigates whether ML can produce representations of LHC collision events that naturally cluster Standard-Model processes and remain sensitive to BSM signatures — and whether that capability follows predictable scaling laws.

- Goal: latent representations of LHC L1 trigger events, useful for both physics clustering and anomaly sensitivity.
- Paradigms in scope: supervised classification (today's evidence) → Autoencoder / BERT-style / VICReg / DINO / JEPA (planned).
- Three research questions anchor everything that follow:
  a. Which learning process yields the optimal representation of LHC collider data?
  b. How does representation quality evolve with model and dataset size?
  c. Can scaling laws quantify anomaly-detection capability with scale?





Dataset, Representation & Workflows

100 L1 PUPPI candidates/event

pT · η · φ · D0 · PID

▶

24 features

after PID one-hot encoding

▶

Linformer / MHA Encoder

ngt_model · PyTorch
MLflow-tracked

▶

Latent (128-d)

cross-entropy head
+ eval_quality

Linformer Notebook

Historical baseline · Exploratory - Manual MLflow logging

ngt_model MHA Package

Scriptable · MLflow-tracked - Latent_dim sweep ready

Sibling project ctl2_model: quantized Linformer → FPGA synthesis at 16 Hz trigger-rate target (hardware-deployment arm, not the focus of this talk).

Evaluation stack

· Standard classifier metrics: accuracy, F1, confusion matrix, ROC-AUC per class

· Composite latent-quality score: alignment loss · participation ratio · GMM gap · Mahalanobis distance

· eval_quality adapted from existing HH_4b-anomaly evaluator (collaborator codebase)

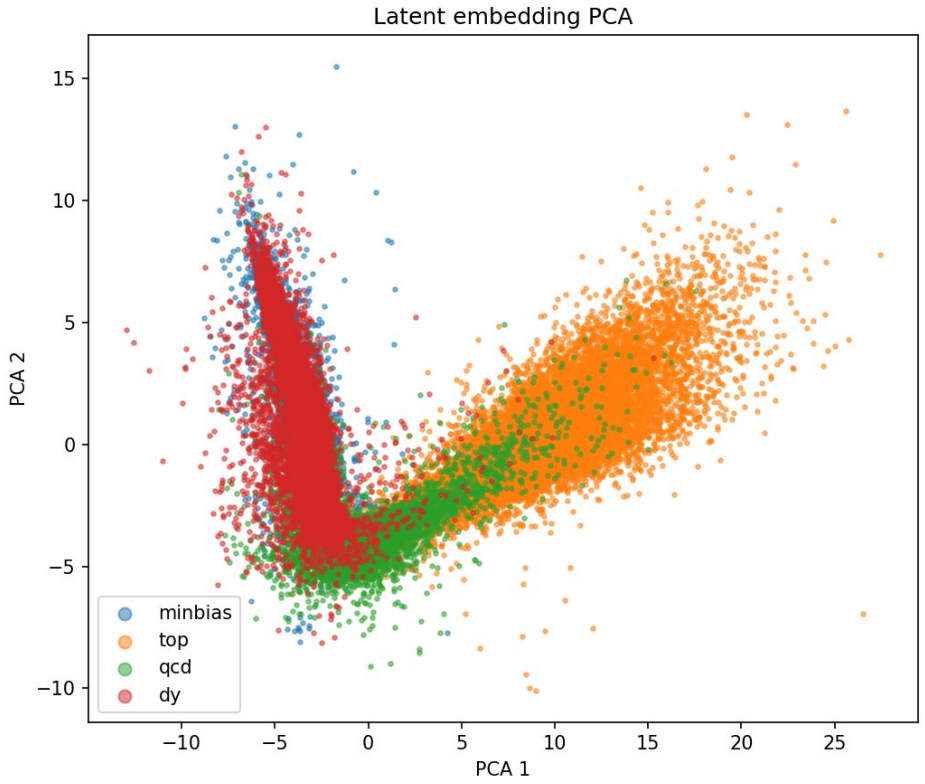
Baseline: Some Processes Separate Cleanly — One Doesn't

↑ Top & QCD separate from MinBias from the start — AUC ≈ 0.94–1.00 on linear probes.

✗ DY ↔ MinBias overlap is representational: linear-probe AUC ≈ 0.54. Not a classifier-head problem.

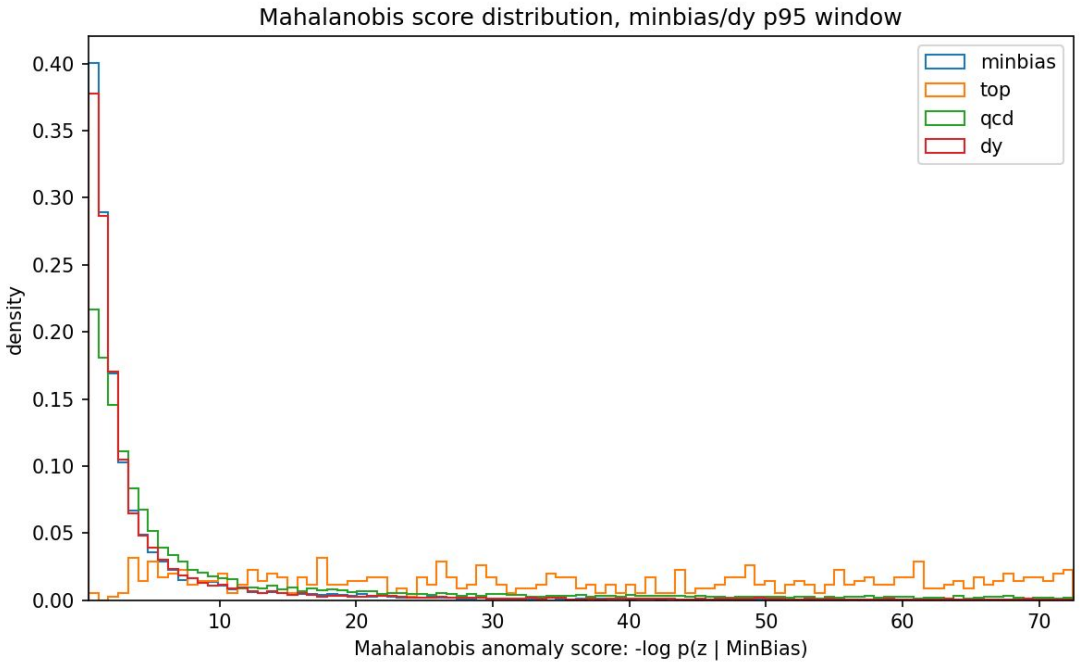
→ 1k→100k events: accuracy 0.63→0.70, but DY→MinBias confusion stays 76–78% either way.

~ Wider attention + augmentation shifted boundary (DY recall 0.56→0.69) but dragged MinBias precision down.

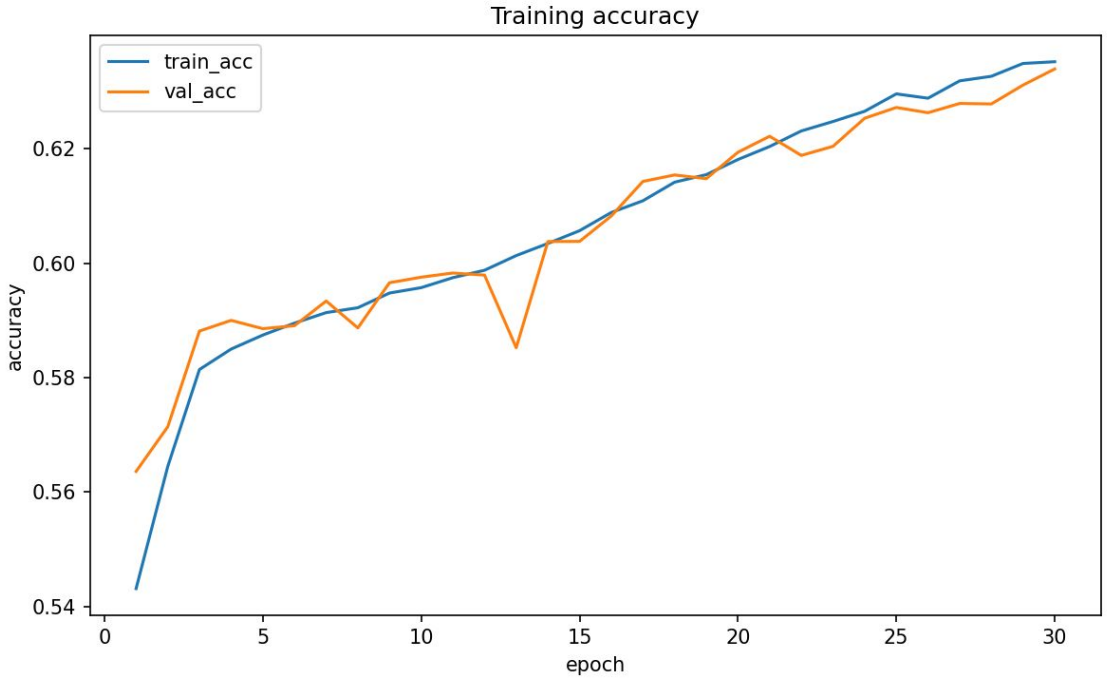


embedding_pca_eval.png — 2026-07-13 MHA run. MinBias (blue) and DY (red) occupy the same region; Top (orange) and QCD (green) separate cleanly to the right.

MHA Baseline: DY-MinBias Separation before HH4B

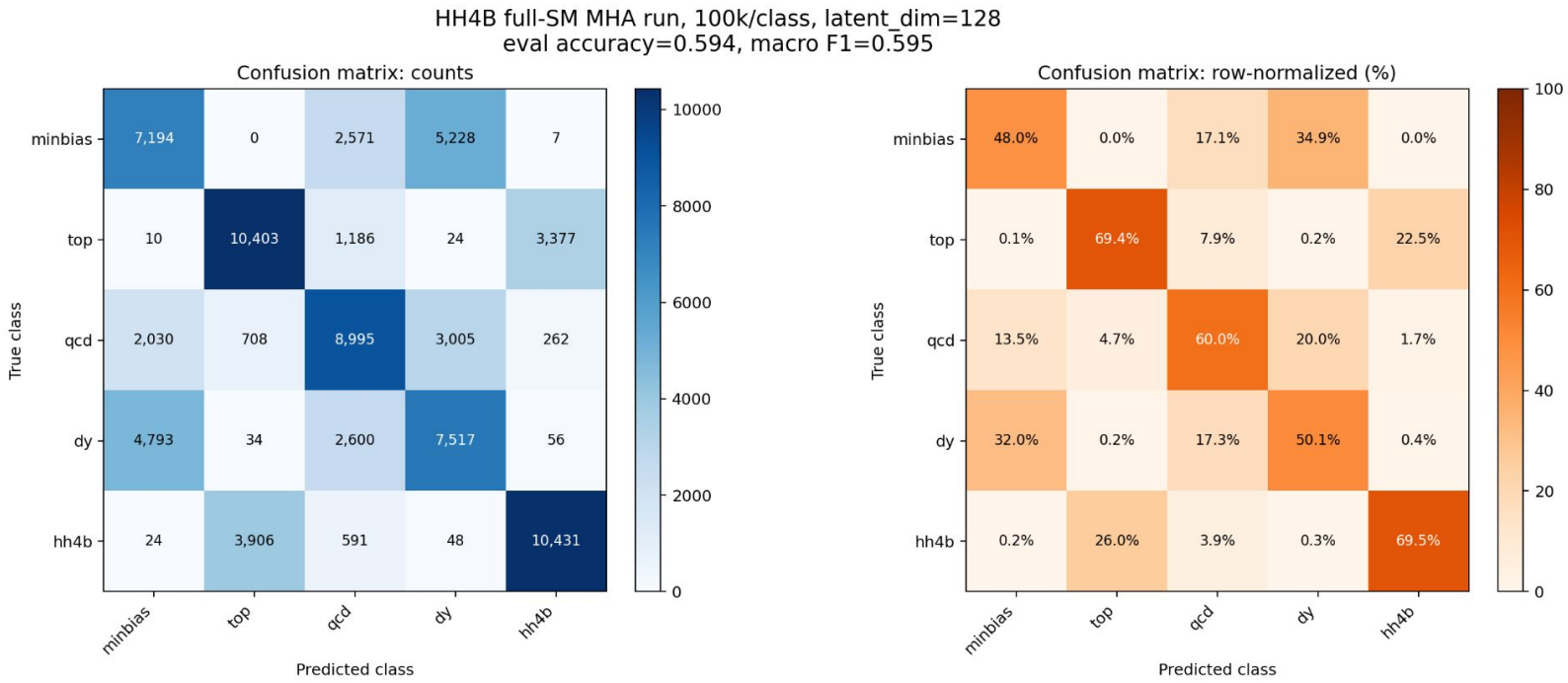


mahalanobis_scores_eval_focus_minbias_dy_p95.png — 2026-07-13 MHA run. MinBias and DY distributions (blue/red) sit almost on top of each other; Top (orange) spreads to much higher scores. A second, independent metric telling the same story as the PCA scatter.



training_accuracy.png — 2026-07-13 MHA run. Train/val accuracy track closely through 30 epochs — stable convergence, no overfitting gap. Representative of the workflow's general training behavior.

HH4B as a Real Anomaly Target



0.594
Overall Accuracy

0.716
HH4B F1 (best class)

26%
HH4B → Top confusion

22.5%
Top → HH4B confusion

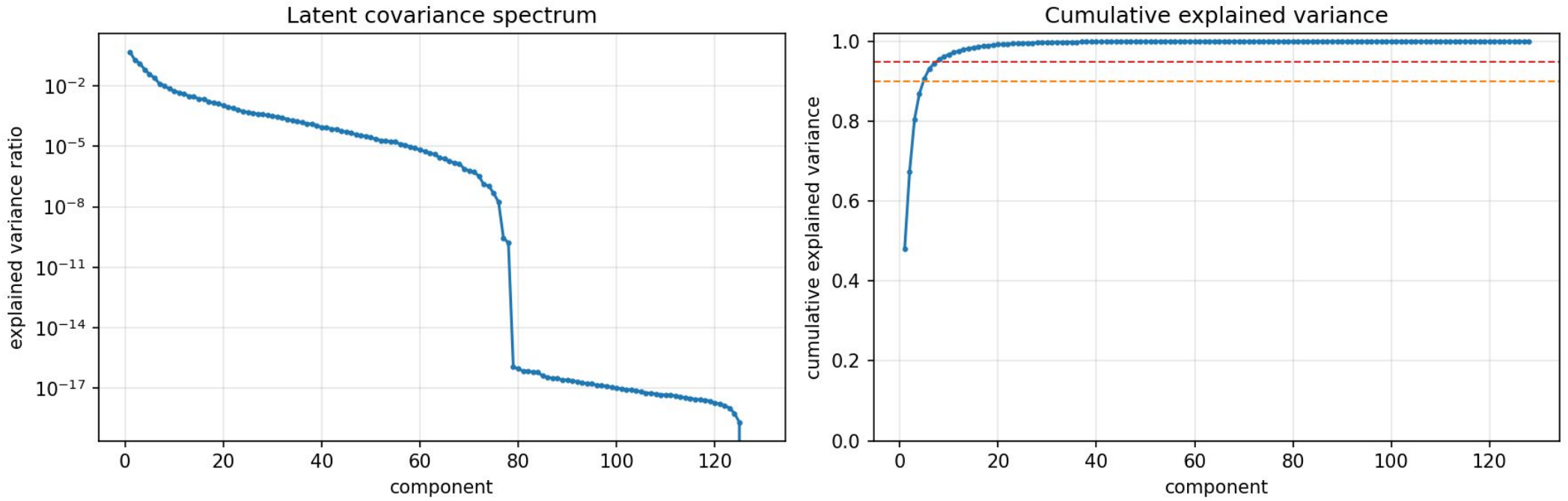
Latent Geometry: How Many Dimensions Are Really Used?

3.42
Participation Ratio

5.49
Entropy Effective Rank (of 128)

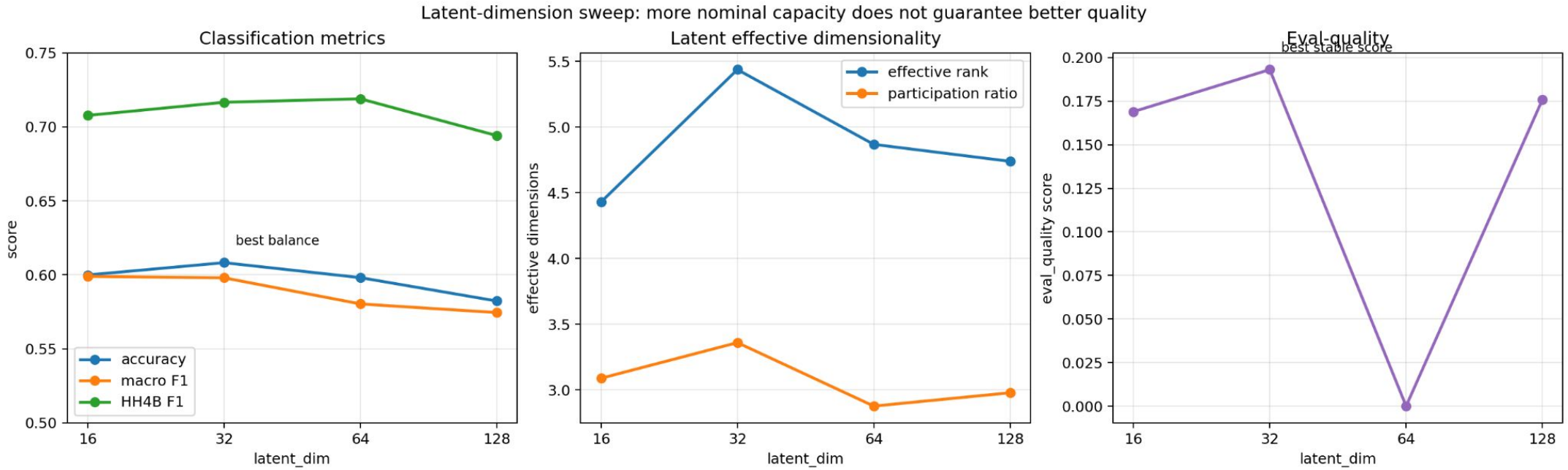
90.7%
Variance in first 5 PCs

~78
Components where eigenvalues cliff to 10^{-17}



latent_effective_dim_spectrum.png — left: log-scale eigenvalue spectrum showing the ~78-component noise cliff; right: cumulative explained variance crossing 90%/95% by components 5/8.

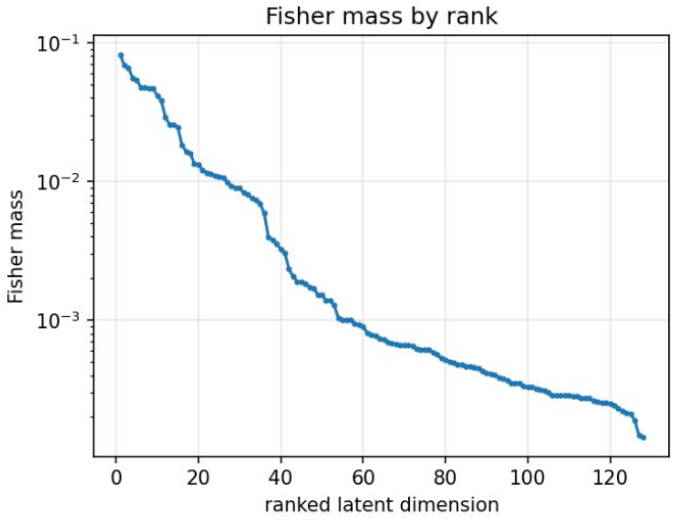
Latent-dimension sweep details

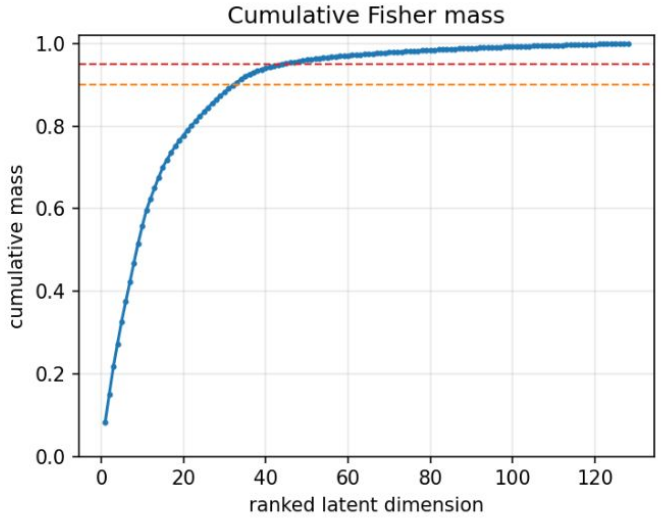


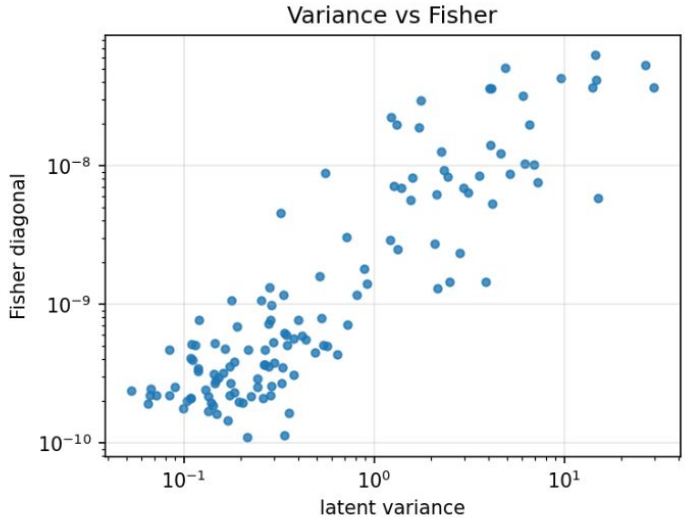
- Fixed setup: HH4B vs full SM, 100k events/class.
- Swept only latent_dim: 16, 32, 64, 128.
- latent_dim=32 gives the best balanced behavior across accuracy, HH4B F1, and eval-quality.

Controlled Sweep: Does More Latent Capacity Help?

latent_dim	Params	Accuracy	Macro F1	DY F1	HH4B F1	Eff. Rank	Part. Ratio
16	181,526	0.600	0.599	0.456	0.708	4.43	3.09
32	185,638	0.608	0.598	0.400	**0.717**	5.44	3.36
64	193,862	0.598	0.580	**0.315**	**0.719**	4.87	2.88
128	210,310	0.582	**0.575**	**0.365**	0.694	4.74	2.98

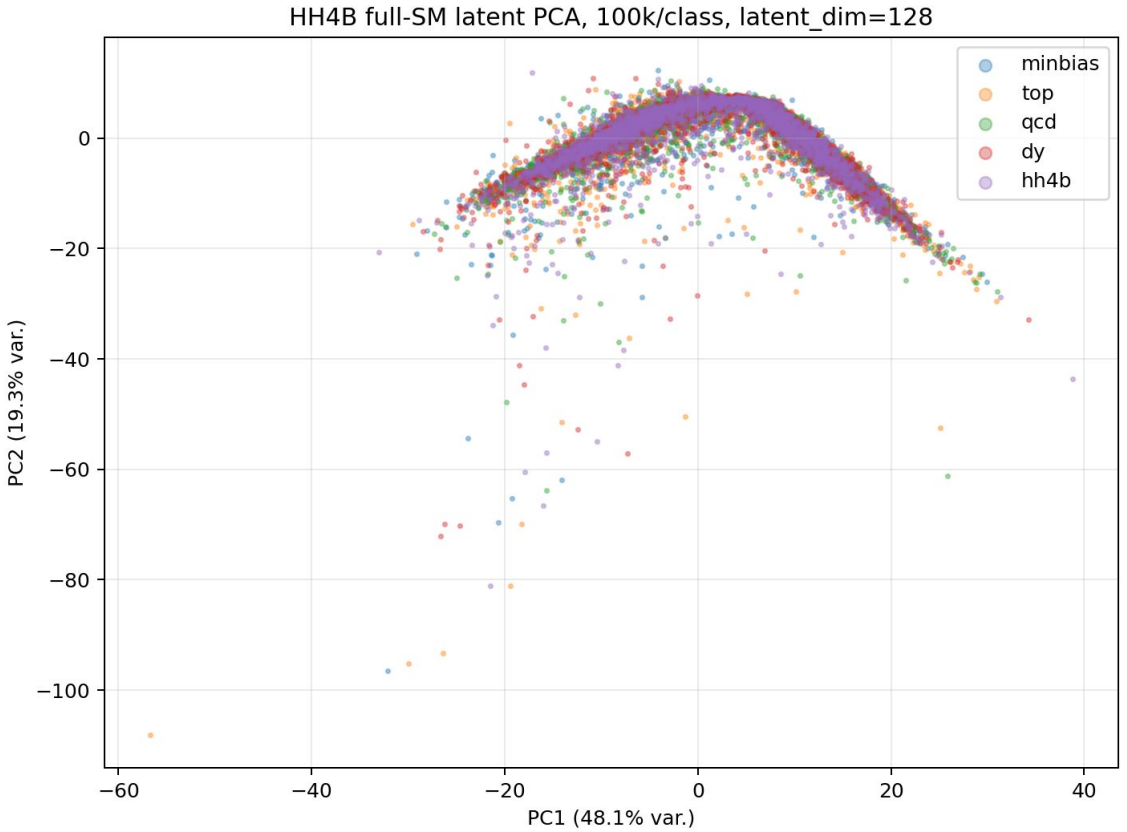






Key limitation

- DY-MinBias overlap remains unresolved in both latent space and classification.
- HH4B is visible as a signal, but it still overlaps significantly with Top.
- Increasing latent_dim alone does not improve performance; latent_dim=32 was more balanced than 64 or 128.
- PCA diagnostics show strong geometric compression of the latent space.
- Fisher diagnostics show the classifier uses more dimensions than PCA suggests, but still far fewer than the nominal 128.
- Different metrics do not always agree, so model selection cannot rely on accuracy or eval-quality alone.



Next Steps

Increase physics-label granularity:

- DY → ee, μμ, ττ
- WW → leptonic, hadronic, semileptonic
- Additional diboson / associated-production classes if available

Separate two experimental axes:

- Fixed granularity + increasing sample size
- Fixed sample size + increased label granularity

Continue latent-space diagnostics:

- PCA / covariance effective rank
- Empirical Fisher in latent space
- Hessian/Fisher approximation on classifier head

Code and analysis repository



Connecting to Scaling Laws (RQ3)

NOT SUFFICIENT ALONE

Latent dimension as scaling axis

- 8× larger latent space did not improve quality
- PCA effective rank remains low (~4–6 geometric directions)
- Fisher shows more task-relevant directions (~30–40), close to latent_dim=32

OPEN QUESTION

Samples : Parameters ratio

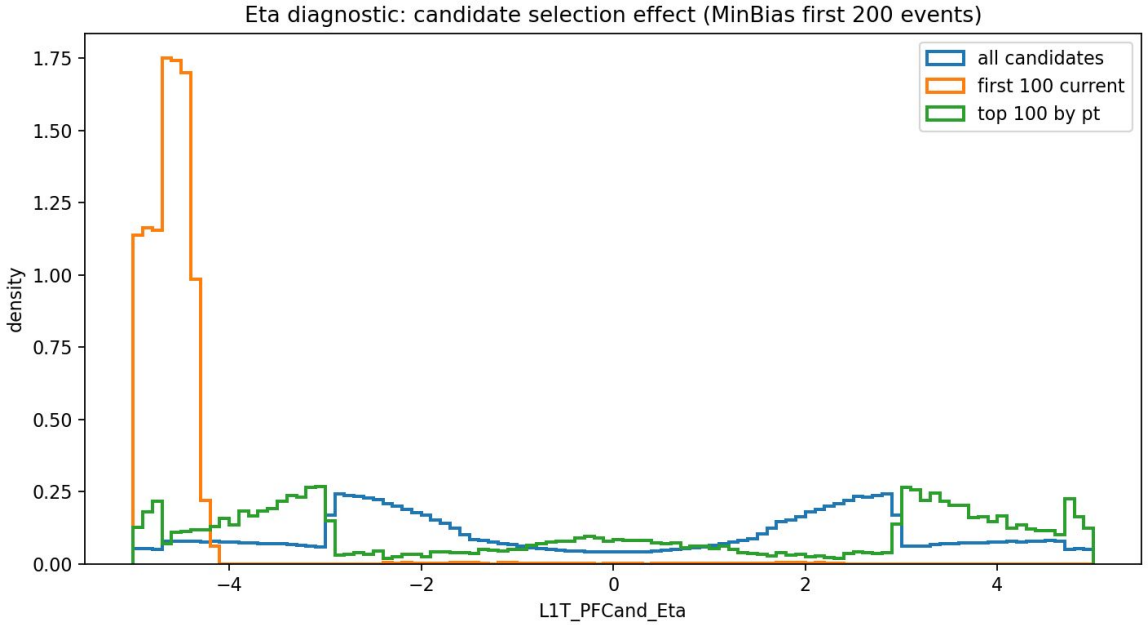
- 350k training events vs 210k params ≈ 1.7:1
- Much lower than Chinchilla ~20:1 heuristic
- No clear overfitting gap observed

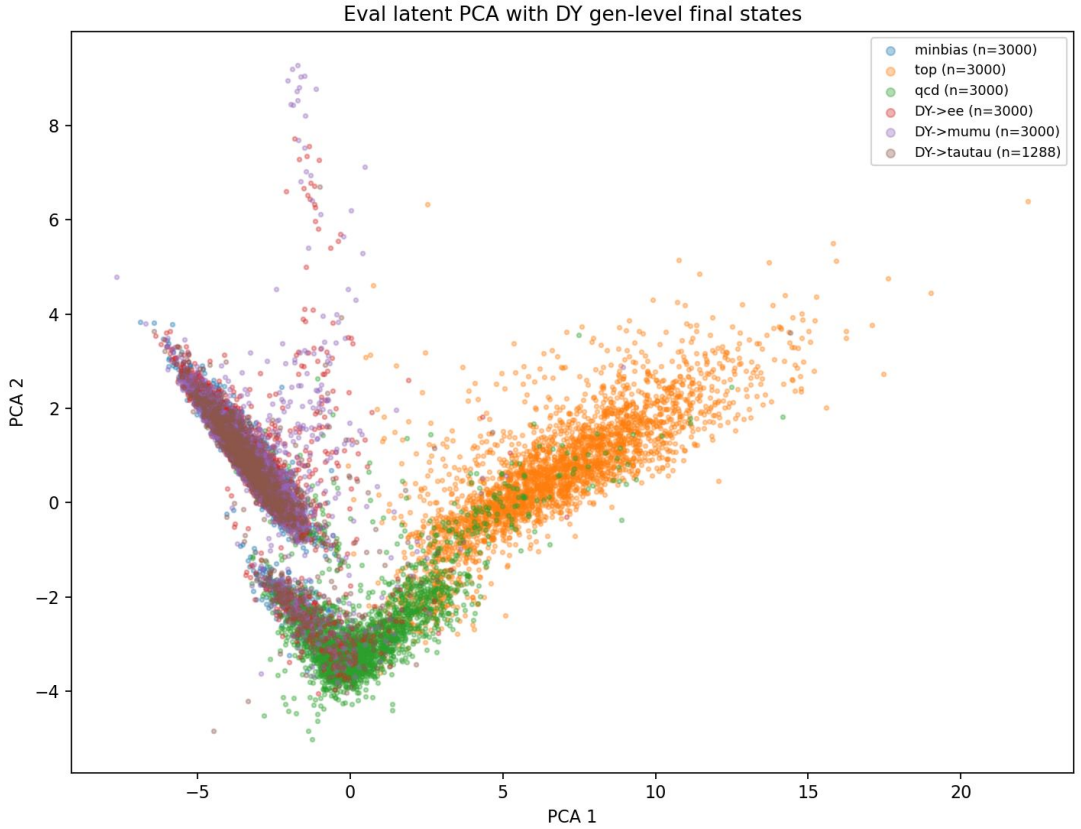
NEXT SCALING AXES

Dataset size & Learning objective

- Increase sample size at fixed granularity
- Increase label granularity at fixed sample size
- Test whether different objectives improve effective latent usage

Backup figures





Backup figures

